PORTFOLIO · FABRICATION & MECHANICAL

JOHN NORTHRUP

Architectural designer working across documentation, fabrication, and code.

I produce permit sets for city and county approval, carry fabrication from concept through install, and build Rhino, Grasshopper, and Python tools that speed up design work.

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FABRICATION & MECHANICAL

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01

ELF-O-MATIC-3000

FA22, MONSTER CITY STUDIOS
ROLE · TECHNICAL DESIGNER & PROJECT MANAGER

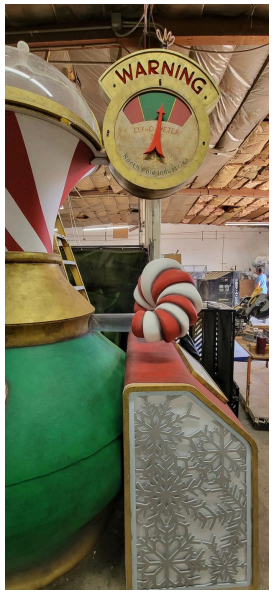
A holiday show piece built at Monster City Studios: a steam-powered candy machine with a neon marquee, a lit viewing window of tumbling balls, a STEAM grille, a pressure gauge and a candy-striped dome. Behind the theming is a working show, with lighting, moving parts and an operator panel to reset it between guests.



BUILT WORK | THE MACHINE RUNNING IN THE SHOP (VIDEO: JCNORTHRUP.COM/PROJECTS/E...



BUILT WORK | MARQUEE, GAUGES AND DOME



DETAIL | WARNING GAUGE, ...



DETAIL | OPERATOR PANEL: ...



DETAIL | SERVICE HATCH WITH LABELED CONTROL SWITCHES



02

DETAILS, DETAILS

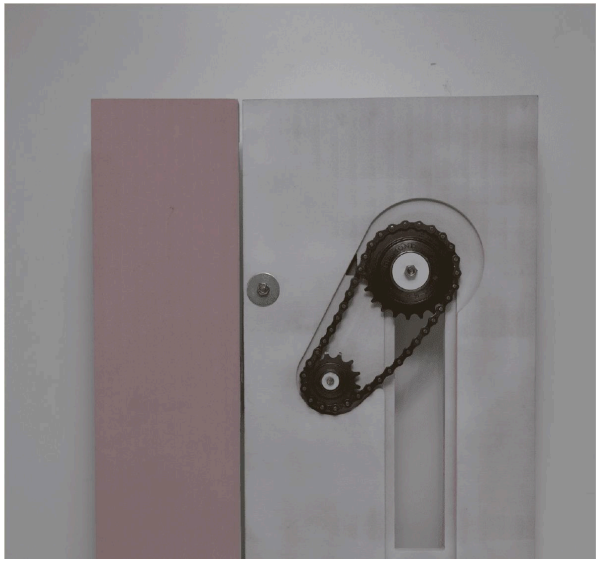
F25, DETAILS, DETAILS
ADVISOR · DWAYNE OYLER

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.

*All work made with partners Chia-Chen Li and Baizhen Yu



FOAM PROTOTYPING PROCESS



PROTOTYPE INTERIOR MECHANISM



FINAL DOOR MECHANISM, CLOSE-UP



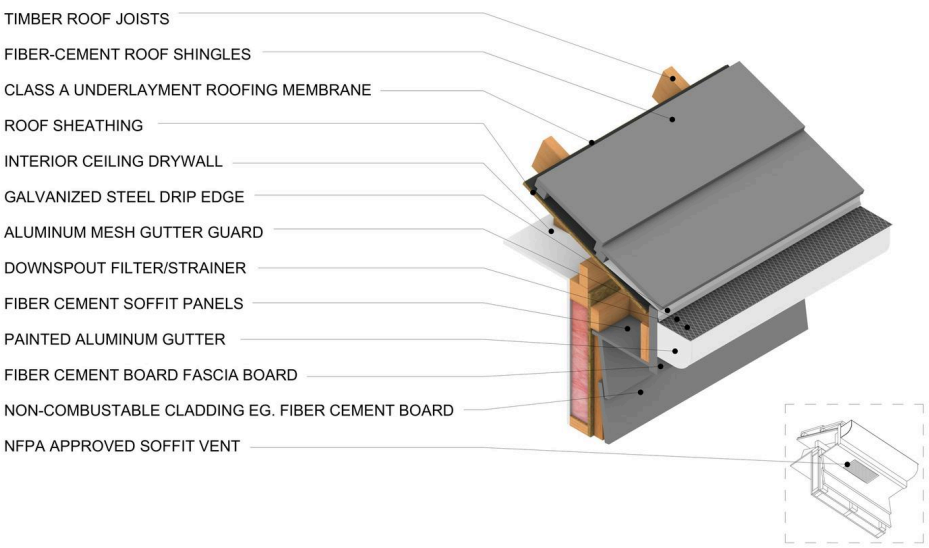
03

RESILIENT FUTURES

SCI-ARC, RESILIENT FUTURES TASK FORCE

A set of wildfire-resilient timber assemblies, drawn as annotated details and then built at full scale. Three conditions were modeled in Rhino and labelled layer by layer: the eaves and overhang, the wall, and the deck. The specification is the argument. Fiber-cement shingles over a Class A underlayment membrane, non-combustible fiber-cement cladding, soffit and fascia, NFPA-approved soffit vents, an aluminum mesh gutter guard and a downspout strainer, fire-rated insulation and a fire-rated window system with tempered insulated glass.

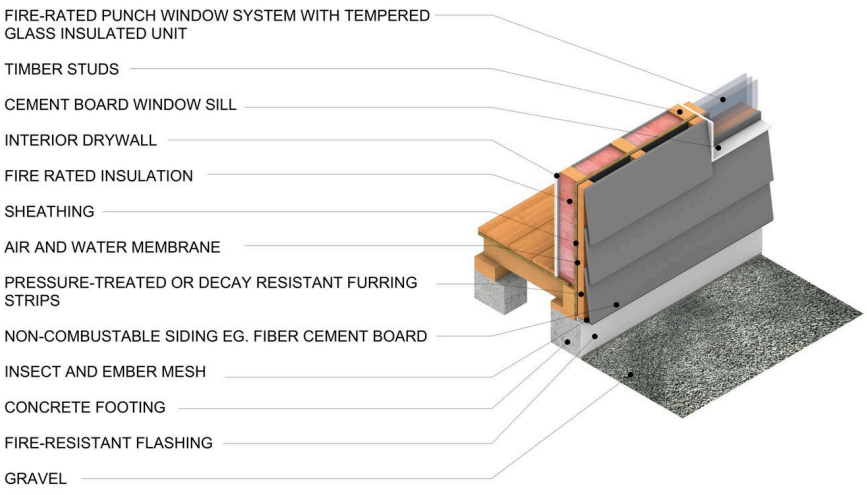
The mockups put the same assembly in the hand at full size: the roof build-up lifted clear of the joists, the vented soffit and guarded gutter seen from below, and the timber frame standing behind the cladding.



DRAWING | EAVES AND OVERHANG, LAYER BY LAYER



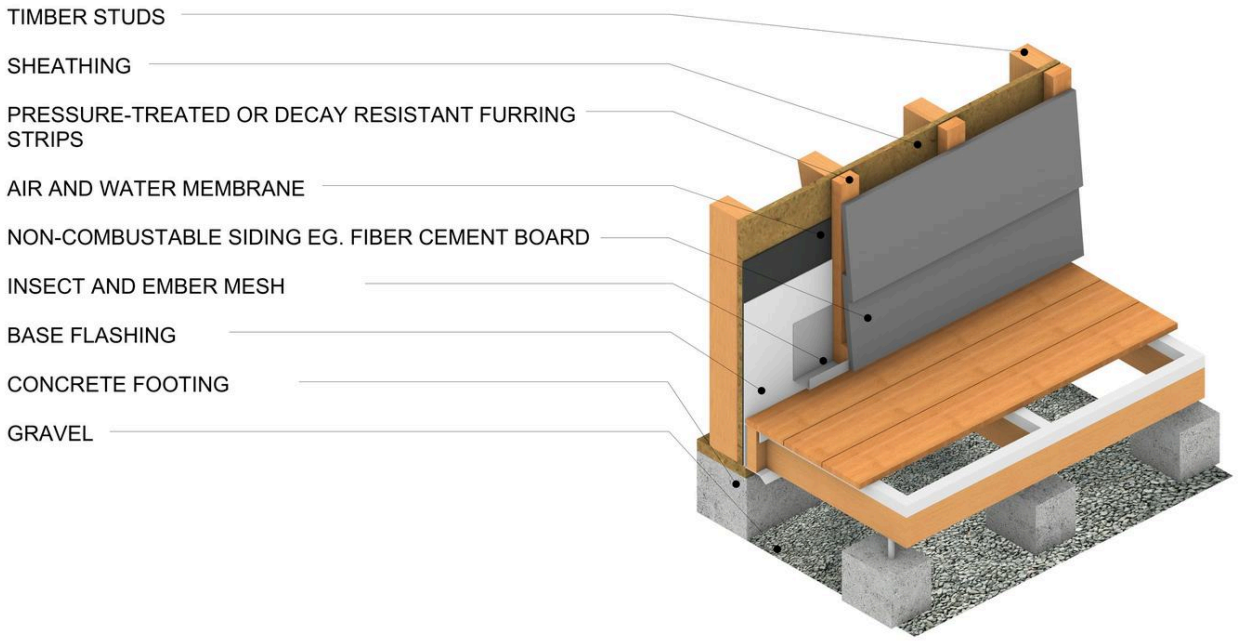
MODEL | THE VENTED SOFFIT AND GUARDED GUTTER, FROM BELOW



DRAWING | WALL, WITH THE FIRE-RATED WINDOW AND SILL



MODEL | THE TIMBER FRAME BEHIND THE CLADDING



DRAWING | DECK, FURRING AND NON-COMBUSTIBLE SIDING



MODEL | THE ASSEMBLY AT FULL SCALE